with 99% confidence. The *Simulator* simulates the placement of these servers on a HS-21 Bladecenter according to the configuration given by the *Arbitrator*. For comparison, we used the following algorithms

- *Load Balanced*: This placement strategy places the VM in a manner such that the load is balanced across all the blades in the Bladecenter.
- *Static*: This algorithm takes long term history into account to find the placement that minimizes the power. The algorithm first estimates the minimum number of servers required to serve all the requests without violating blade server capacities. It then places the VMs (only once) on the servers to minimize power
- *mPP*: The minPowerPlacement Algorithm dynamically determines the placement that minimizes the power for that time window and is based on FFD.
- *mPPH*: The minPowerPlacement with History Algorithm determines the power-minimizing placement that takes into account the previous placement.
- *PMaP*: The PMaP Algorithm optimizes the tradeoff between power cost and migration cost, while computing the new placement for the window.

In our experiments, the *Performance Simulator* maps each server utilization trace to a Virtual Machine (VM) on the HS-21 with the VM size being set to the CPU utilization on the server. The trace is divided into time windows, and in each time window, the *Arbitrator* determined a new placement for each of the placement strategies based on the utilization specified by the *Performance Manager* in the time window. We feed the placements to the *Simulator*, which estimates the cost of each placement and logs it.

We studied the various methodologies with respect to the power consumed, the migration cost incurred, and the sum of power cost and migration cost. We then conducted a comparative study of the algorithms with change in server utilization. We also increased the number of blades in the Bladecenter to investigate the ability of the algorithms to scale and deal with fragmentation. We used a metric to quantify the relative impact of migration cost with power cost and termed it *MP Ratio*. The migration cost is determined by estimating the impact of migration on application throughput and consequent revenue loss, computed as per the SLA. This cost is compared with the power cost using the power drawn and the price/watt paid by the enterprise. We pick a reasonable value for this ratio in our baseline setting based on typical SLAs and then vary this ratio to study its impact on the performance of various algorithms. We also plugged in some other power models to investigate if the algorithms are dependant on specific power models. Further, by mixing different kinds of physical servers (with different power models), we investigate the ability of the algorithms to handle heterogeneity. We finally studied the practicality of the assumptions made.

### 5.3 Results

We first study the behaviour of various algorithms as the traces are played with time (Fig. 7(a)). The aggregate utilization of the VMs varies during the run and the dynamic algorithms continually try to adapt to the changed workload. We